

## Trainings ▾

## Select Service Tools

### Recommended Cummins® Service Tools

- No recommended Cummins® service tools

### Additional Service Items

- No additional service items required

## Preparatory Steps

### WARNING

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

### WARNING

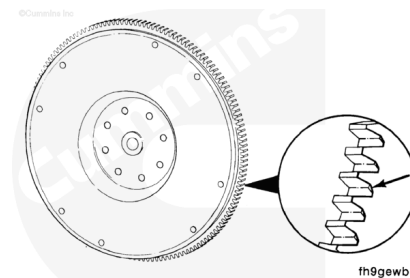
**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the transmission and related components. See equipment manufacturer service information.

## Initial Check

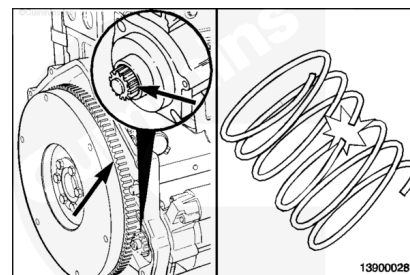
Inspect the flexplate ring gear teeth for damage.

If the flexplate ring gear is damaged, make sure to evaluate the following possible causes prior to replacing the flexplate.



## Mechanical

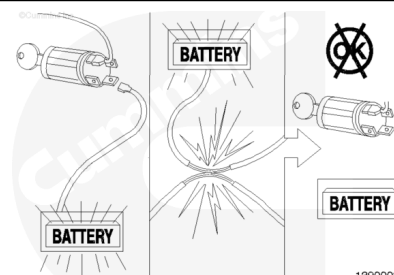
A mechanical issue can typically be identified by seeing damage to the ring gear of the flexplate in 3 distinct locations, commonly called 120 degree milling. The following can be causes for mechanical issues:



1. The possibility of improper starting motor spacing. Refer to Procedure 013-020 in Section 13.  
(/qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html)
2. The interference between the ring gear land area and the starting motor pinion. The wrong starting motor can be installed. See equipment manufacturer service information.
3. The possibility of a problem with the starting motor pinion. Inspect the pinion for nicks and burrs. Refer to Procedure 013-020 in Section 13.  
(/qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html)
4. The torque converter/transmission is damaged or incorrectly mounted. See equipment manufacturer service information.
5. Incorrect starting motor pinion to flexplate ring gear pitch and teeth match. See equipment manufacturer service information.

## Electrical

An electrical issue can typically be identified by seeing damage to the ring gear of the flywheel 360 degrees around the circumference of the ring gear (commonly called 360 degree milling). The following can be causes for electrical issues:



1. The operator is attempting to start the engine while the engine is already running. Check if a Starter Lockout feature is available through the original equipment

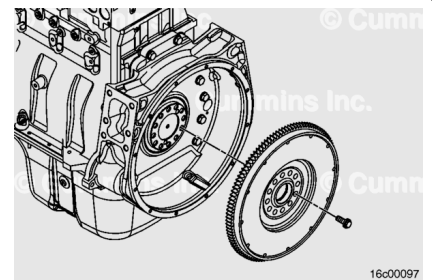
manufacturer (OEM) (activated with INSITE™ electronic service tool) or the starting motor manufacturer.

2. The keyswitch is causing intermittent starting motor engagement when the engine is running. Inspect the keyswitch. Refer to Procedure 013-018 in Section 13. (</qs3/pubsys2/xml/en/procedures/377/377-013-018.html>)
3. The orientation of the starter relay, so that the direction of the pull contact is in the direction of the vehicle's travel. This results in intermittent starting motor engagement when the engine is running. Relocate the starter relay. See equipment manufacturer service information.
4. Intermittent starting motor wiring issues. See equipment manufacturer service information.

## Remove

Remove the flexplate capscrews and flexplate.

**Note :** Some flexplates require mounting plates and/or adapters. It will be necessary to remove any mounting plates and/or adapters prior to or with the flexplate. Make sure to note the location of any mounting plates and/or adapters for later installation.

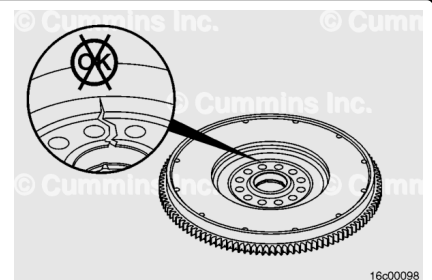


## Inspect for Reuse

Inspect the teeth of the ring gear for chips or uneven wear.

Check the flexplate for cracks.

Replace the flexplate if any damage is found.



# Install

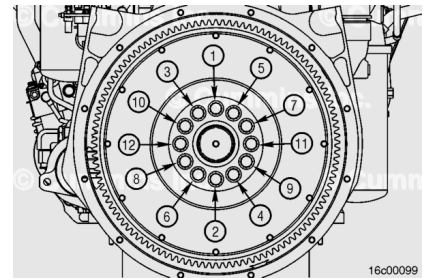
**Note :** Some flexplates require mounting plates and/or clamp rings. It will be necessary to install any mounting plates and/or clamp rings prior to or with the flexplate as noted during removal.

Install the flexplate capscrews and flexplate.

Tighten the capscrews in the sequence illustrated.

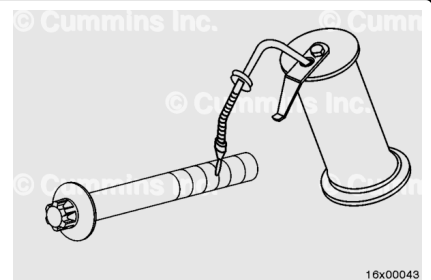
**Torque Value:** 298 n·m [ 220 ft-lb ]

Tighten the capscrews again in the sequence illustrated.



**Note :** Some flexplates require mounting plates and/or clamp rings. Install any mounting plates and/or clamp rings prior to or with the flexplate as noted during removal.

Lubricate the threads of the capscrews and surface of the built-in washers. Use clean lubricating engine oil. Wipe off excess oil with a clean cloth.



## Finishing Steps

### **⚠ WARNING ⚠**

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- Install the transmission and related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.

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**Last Modified: 19-Aug-2025**

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